

14 Catalytic Rate Laws

Why are we learning this?

So far, the rate laws in this course have primarily been written as simple functions of concentration or pressure. Catalytic reactions behave differently because reactions occur on catalyst surfaces. Before a molecule can react, it often must first adsorb onto a catalyst site. As a result, catalytic reaction rates depend not only on reactant concentration, but also on the availability of catalyst sites. This influences the mathematical forms of the rate laws.

By the end of today's lesson, you should understand:

- Why catalytic rate laws differ from simple power-law rate laws.
- What catalyst sites are.
- What adsorption means.
- What surface coverage means.
- Why catalyst sites are a limited resource.
- How adsorption influences reaction rate.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws

(Kinetics)



Equilibrium

(Reversible Reactions)



Equilibrium Conversion

(Thermodynamic Limitations)



One Reaction: Solving in terms of Reaction Progress



One Reaction: Reactor Design

(CSTR, Batch, PFR, PBR)



► Catalytic Reactions ◀



Numerical Methods

(Python)



Energy Balances



One Reaction: Non-Isothermal Reactor Design



Multiple Reactions



Reactor Safety



Challenging the Well-Mixed Assumption

(Residence Time Distributions)



Challenging the Elementary Rate Law Assumption

(Reaction Mechanisms)

Big Idea

Reaction rate depends on: Reactant availability + Catalyst-site availability.
Catalytic surfaces are a limited resource.

The reaction $2A \rightarrow B$ is carried out in the gas phase over a catalyst. The rate law is:

$$-r_A = \frac{kP_A^2}{(1 + K_A P_A + K_B P_B)^2}$$

Where $k = 1250 \frac{\text{mol}}{\text{atm}^2 \cdot \text{L} \cdot \text{s}}$, $K_A = 4 \text{ atm}^{-1}$, and $K_B = 13 \text{ atm}^{-1}$.

What is different about this rate law, compared to the ones we've been working with?

1. Rate given in terms of

1. There is an equation

Why Does This Rate Law Look Different?

Most previous rate laws were written as simple power-law expressions. Catalytic rate laws often contain additional terms because reaction occurs on catalyst surfaces rather than in the bulk fluid. The rate therefore depends not only on reactant concentration but also on the availability of catalyst sites.

On a catalyst, a "site" is

An example of a catalyst site is a Pt atom on a slab of Pt metal. For example, the Pt metal atom will easily dissociate the O_2 molecule, and hence Pt catalysts are commonly used to promote oxidation reactions.

The denominator is a "coverage term". It tells us

What Is a Catalyst Site?

A catalyst site is a location on the catalyst surface where adsorption and reaction can occur. Examples include:

- Metal atoms on a catalyst surface.
- Pockets within an enzyme.
- Acid sites in zeolites.

If no site is available, the catalytic reaction often cannot proceed.

Catalyst Sites Are Limited

A catalyst contains only a finite number of active sites. As molecules adsorb:

Occupied Sites Increase



Free Sites Decrease



Additional Adsorption and Reaction Become Harder

Competition for sites is one of the defining features of catalytic kinetics.

What Does the Coverage Term Represent?

The coverage term accounts for how catalyst sites are distributed among adsorbed species. As more sites become occupied:

Fewer Sites Available



Lower Capacity for Additional Adsorption and Reaction



Potentially Lower Reaction Rate

The denominator captures this competition for catalyst surface area.

We will explore this in more detail later, but for now, let's pause and illustrate coverage: [links to powerpoint in OneDrive](#)

The free site coverage is denoted

θ_* is the fraction of catalyst sites

In our example, $\theta_* =$

So, now we need expressions for θ_A and θ_B .

We get these by considering the adsorption of these species on the catalyst.

What Is Surface Coverage?

Surface coverage measures the fraction of sites occupied by a particular species.

Examples:

θ_A = fraction occupied by A

θ_B = fraction occupied by B

θ_* = fraction of free sites

All catalyst sites must be accounted for.

Why Are Free Sites Important?

Many catalytic mechanisms require at least one vacant site before adsorption can occur. If most sites are occupied:

Adsorption Slows



Reaction May Slow

Free-site coverage is therefore a critical variable in catalytic reaction engineering.

Adsorption occurs when

We assume that A (g) and B (g) are catalyst

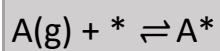
with the

Hence, we can write $K_A =$

and $K_B =$

What Is Adsorption?

Adsorption occurs when a molecule binds to a catalyst surface. For example:



where:

* = free site

A* = adsorbed A

Adsorption is often the first step in a catalytic reaction mechanism.

Hence, θ_* =

Our rate law is $-r_A = \frac{kP_A^2}{(1+K_AP_A+K_BP_B)^2} =$

Why Solve for Surface Coverage?

The reaction rate depends on the number of available catalytic sites. The number of available sites depends on:

- Adsorption of A.
- Adsorption of B.
- The fraction of free sites.

To predict reaction rates, we must be able to quantify all three.

Adsorption Has Competing Effects

Adsorption is beneficial because reactants typically must adsorb before reacting. However, excessive adsorption can occupy too many catalyst sites.

Too Little Adsorption



Few Reacting Molecules

Too Much Adsorption



Too Few Free Sites

Many catalytic systems operate most effectively at intermediate coverages.

How do the coverages of A and B influence the rate? Why?

Summary and Key Takeaways

Today we introduced the concepts behind catalytic rate laws.

Key Ideas

- ✓ Catalytic rate laws often differ significantly from simple power-law expressions.
- ✓ Catalytic reactions occur on catalyst sites.
- ✓ Catalyst sites are a limited resource.
- ✓ Surface coverage describes how catalyst sites are distributed among adsorbed species as well as how many sites are available to carry out the reaction.
- ✓ Free-site coverage strongly influences catalytic reaction rate.
- ✓ Adsorption controls catalyst-site availability.
- ✓ High surface coverage can sometimes inhibit reaction by blocking catalyst sites.

Most Important Idea

Reaction Rate



Depends on



Available Catalyst Sites



Which Depend On



Adsorption and Coverage

Understanding surface coverage is the foundation for understanding catalytic mechanisms and catalytic reactor performance.